

Software Testing Plan

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CSE General

1. Introduction

This document outlines the overall testing strategy, scope, and objectives for the E-Book Management System.

The goal of testing is to ensure that the system functions correctly, meets user requirements, and is free from critical defects before deployment.

2. Test Strategy

The following testing levels are used in the project:

- **Unit Testing:**
Individual modules such as login, book upload, search, and user management are tested independently.
- **Integration Testing:**
Interaction between modules (frontend, backend, and database) is tested to ensure smooth data flow.
- **System Testing:**
The complete E-Book Management System is tested as a whole to verify functional and non-functional requirements.
- **Acceptance Testing:**
Final testing is performed to confirm that the system meets user and project requirements.

3. Test Environment

The test environment specifies the resources required for testing.

- **Hardware:** Personal computer or laptop
- **Software:** Web browser, database server, application server
- **Test Data:** Sample user accounts, book records, and search queries

4. Test Cases

Test cases are designed to validate system functionality.

Each test case includes:

- Input data
- Expected output
- Actual output
- Pass/Fail criteria

This ensures systematic and repeatable testing.

5. Defect Management

Defect Management defines the process for handling errors found during testing.

- Defects are logged with details such as description and severity
- Developers analyse and fix reported defects
- Retesting is performed after fixes
- Ensures improved software quality and reliability